

NurdleDNA: An Edge-AI and Smart IoT Inline System for Source-Level Microplastic Detection

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Abstract— Coastal industrial water infrastructure requires prompt capture of microplastics from pollution sources to intercept microplastics prior to leaving the discharge points and entering the ecosystem. Pre-production plastic pellets (nurdles) pose unique challenges for management because of their size, mobility, similarity to other suspended particles, and handling bulk through petrochemical facilities, ports, and industrial storm drains. Existing techniques such as visual inspection, passive filters, turbidimeters, and FTIR and Raman analysis in the lab cannot work in real-time or do not offer source-level containment and feedback on the discharge streams. In this paper we introduce NurdleDNA – an edge-AI and IoT-based system that detects microplastics, isolates the stream, collects physical evidence, and logs every occurrence into the cloud. The system utilizes optical vision chamber, detection using YOLOv8, LDR turbidity sensor validation, MQ-135 gas sensor, load cell-based capturing measure, and servo-controlled pinch valve into a closed loop detect-classify-actuate-capture-log framework. Probabilistic edge-AI inference on the NVIDIA Jetson Nano device is separated from deterministic finite state machine control on the Arduino Uno device for physical containment fail-operational mode. Bench-scale prototype testing demonstrates contamination detection, automatic flow isolation within ~300 ms of an alarm, gravimetric evidence captures, and time-stamped audit logging, with an end-to-end response time of approximately 3 s.

Keywords— *microplastic detection, nurdles, edge AI, Internet of Things, industrial water monitoring, source-level containment, multi-sensor fusion, cloud audit logging*

I. INTRODUCTION

Plastic pollution has emerged as one of the most pressing environmental challenges of the modern era. Among the most concerning contributors are nurdles, which are pre-production plastic pellets measuring 3–5 mm in diameter, released into aquatic systems through polymer manufacturing plants, shipping ports, and logistics facilities via drainage networks. Once in the environment, nurdles and other microplastics (plastics less than 5 mm in diameter) persist in aquatic systems, progressively fragmenting into smaller particles that

accumulate in ecologically sensitive areas and enter food chains. Research conducted in Dubai and across the broader MENA (Middle East and North Africa) region underscores the urgency of monitoring nurdle and microplastic pollution in both coastal and industrial environments [1].

The environmental consequences of this contamination extend beyond immediate ecological harm. Microplastic pollution degrades natural habitats and disrupts ecological cycles, thereby reducing ecosystem resilience against climate change. Industrial drainage pathways compound this problem further, as they can simultaneously carry nurdles alongside volatile organic compounds (VOCs) and hydrocarbon contaminants, creating a multi-hazard environment that conventional passive filters and manual monitoring approaches cannot adequately address. The small size of nurdles, their physical homogeneity, and their high mobility through drainage networks make source attribution exceptionally difficult, presenting a significant obstacle to effective pollution control.

To address this challenge, this paper introduces NurdleDNA, an inline IoT-based monitoring and capture system designed to detect nurdles in industrial water flows in real time. The system employs a camera-based AI vision layer running on an NVIDIA Jetson Nano, using a YOLOv8-based computer vision pipeline for nurdle detection [2], supported by UV illumination and an LDR turbidity sensor. A parallel gas sensing subsystem monitors for VOC accumulation, while a servo-controlled diverter valve automatically isolates flow upon detection. All contamination events are logged and transmitted via Firebase to a remote cloud dashboard for real-time monitoring and compliance reporting.

Whereas most recent contributions have focused on novel object detection architectures or sensing primitives, the primary innovation in this work is a closed-loop edge-AI system as opposed to these individual components. Namely: (1) an end-to-end detection-classification-actuation-capture-reporting system wherein edge-AI vision is combined with a deterministic finite state machine implemented on an Arduino

platform such that physical flow isolation operates fail-operational in case of failure of the edge-AI layer; (2) a multi-sensor fusion mechanism of optical turbidity, VOC, and AI vision sensors with temporal confirmation which prevents optical sensor false positives prior to activation; and (3) an audit trail consisting of three channels of forensic data - detection frame rendered, gravimetric capture mass collected, and VOC logs generated - written immutably to Firebase on every alert. To the best of our knowledge, such an end-to-end inline detection, automated containment, physical capture, and audit logging system has not been shown in an edge device before for nurdle pollution.

This paper is organized as follows: Section II discusses previous works, Section III states the methodology, Section IV covers the results and discussion and Section V oversees the conclusion and future work.

II. LITERATURE REVIEW

A. Machine Learning and Deep Learning for Microplastic Detection

The detection of microplastics is becoming more oriented towards the use of automated machine learning and computer vision techniques, rather than manual laboratory-based procedures. While FTIR, Raman Spectroscopy, microscopy, and visual sorting can give accurate results, these techniques are usually time-consuming and hard to implement on an ongoing basis [3]. It has been demonstrated recently that using artificial intelligence in such applications can increase speed and efficiency, while CNNs and transfer learning can help in classifying microplastics based on images [4]. Nonetheless, these techniques address only detection and classification problems.

B. Climate-Ecosystem Impact and Portable Sensor Monitoring

The problem of microplastics pollution influences ecosystems, biodiversity, and climatic resilience due to habitat disruption, biological processes disruption, and disruption of the ocean's capacity for carbon sequestration [5]. Since delayed laboratory testing is not capable of preventing pollution spread, recent research has focused on portable optical sensing as an alternative fast microplastics monitoring tool [6], [7]. Nevertheless, despite the fact that portable sensing mostly helps with detection and screening, there is a need for inline monitoring of microplastics in industry.

C. Edge-AI, Multi-Sensor Fusion, and IoT Monitoring

Edge-AI and IoT sensing enable real-time microplastic monitoring in industrial water systems. The YOLO family was originally introduced as a unified real-time object-detection framework [8], while recent YOLOv8-based research has demonstrated plastic detection in underwater environments under variable lighting, turbidity, water clarity, and occlusion conditions [9]. In NurdleDNA, YOLOv8-based inference is deployed on the NVIDIA Jetson platform to perform local detection near the sensor with reduced cloud dependency. The system further integrates turbidity, gas/VOC sensing, load-cell measurement, Arduino-based control, and cloud logging to improve detection confidence and support real-time environmental response. This is directly relevant to the NurdleDNA flow-cell environment, where bubbles, glare, lighting variation, and suspended particles may interfere with direct visual classification.

D. Grey Areas and Contribution of This Work

Some of the recent research works have advanced the detection of microplastics through automated quantification, deep learning-based classification, optical sensing, YOLOv8 for Aquatic Plastic Detection, and edge-AI application [6]–[10]. But these are still largely constrained to only detection and classification without inline isolation of flows, proof-of-capture, and audit logging. To fill this void, NurdleDNA comes up with a closed loop system which comprises AI vision, multi-sensor validation, valve actuation, measured mass capture, and immutable cloud audit logging.

III. METHODOLOGY

A. System Architecture and Physical Partitioning

Figure 1 visualizes the NurdleDNA prototype, which implements the dual-layer containment strategy modeled by Autodesk Fusion 360 that insulates the liquid-conducting pathways from the electronics processing the signals.

The wet pathway, including sensor elements, consists of an optical flow cell made of transparent materials and a WS2812B-8 LED strip illuminating the optical flow cell homogeneously. There is also a camera viewport opposite the LED strip. LDR is used as a turbidity sensor. Volatile organic compound (VOC) detection is performed using MQ-135 sensor situated in the headspace. There is also evidence that cartridge suspended from a load cell and a pinch valve actuated by the MG996R servo controlling the isolation of flow. In the dry pathway, including electronic components, there is NVIDIA Jetson Nano with the CNN inference and cloud publishing and Arduino Uno boards executing the deterministic logic.

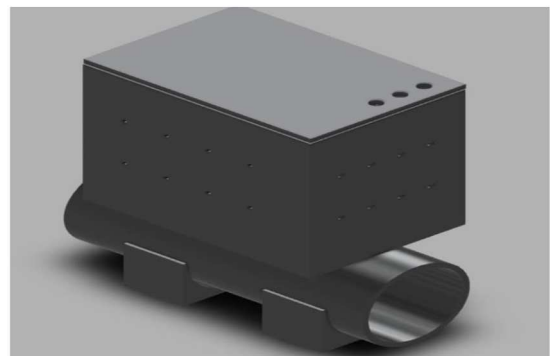


Figure 1: NurdleDNA CAD prototype with electronics enclosure and inline flow conduit.

B. Multi-Sensor Acquisition and Signal Conditioning

Signal conditioning and telemetry are performed by the Arduino Uno at a 5 Hz sample rate. Optoelectronic turbidity is read on pin A0; values exceeding the warning limit ($V_o > 600$) indicate possible contamination. Petrochemical hydrocarbon volatiles are detected on pin A1 via an MQ-135, with a critical alarm raised when ADC values exceed 200. Gravimetric accumulation of spills in the downstream cartridge is measured with a parallel-beam load cell and 24-bit HX711 conversion on pins D4 and D10.

The architecture of the system and its physical implementation are provided in Figures 2 and 3.

temporality, turbidity, and gas sensors must also all be in agreement to trigger actuation, hence NurdleDNA can accommodate an inaccurate detector and does not rely on a single-frame high mAP (Section IV-C). Air bubbles play an essential role operationally since they are the leading source of optical false positives in flow cells (Figure 7); the detector’s relatively high accuracy at detecting bubbles allows it to separate air from actual particle detections. Figure 6 illustrates that the detector performs well for air bubbles and pen classes but struggles with minority particle classes because of severe dataset imbalance and Figure 8 summarizes the operational detection performance of the hybrid YOLOv8–OpenCV pipeline.

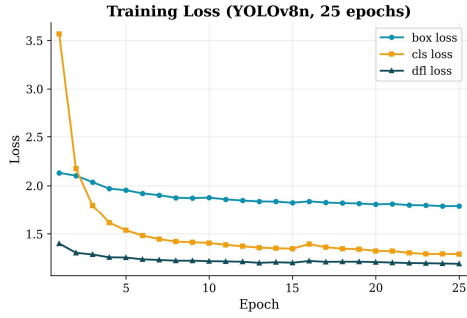


Figure 4: YOLOv8n training loss (box, cls, dfl) over 25 epochs on the Roboflow microplastics-t0ddd v6 dataset (2,171 training images, 19 classes).

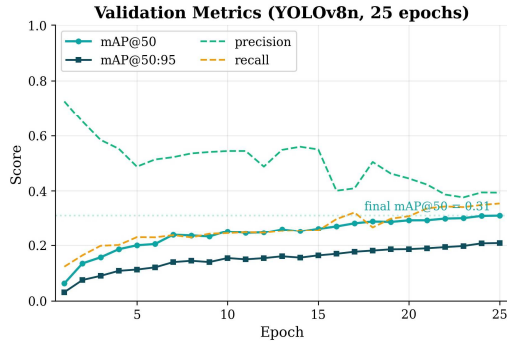


Figure 5: YOLOv8n validation metrics over 25 epochs on the Roboflow microplastics-t0ddd v6 dataset; final mAP@50 = 0.31 (mAP@[50:95] = 0.21).

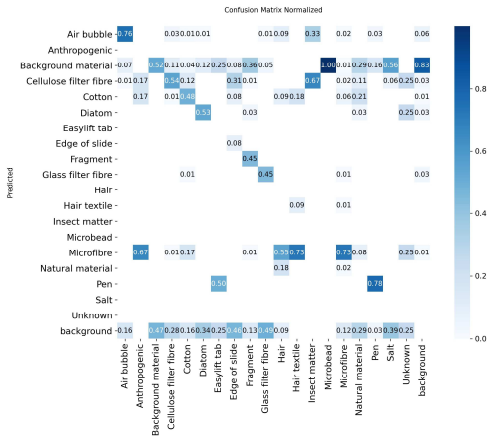


Figure 6: Normalised confusion matrix for the full 19-class public dataset. Air bubble and Pen are recovered most reliably, while many particle classes are absorbed into the dominant “background material” class, reflecting the dataset’s severe imbalance.

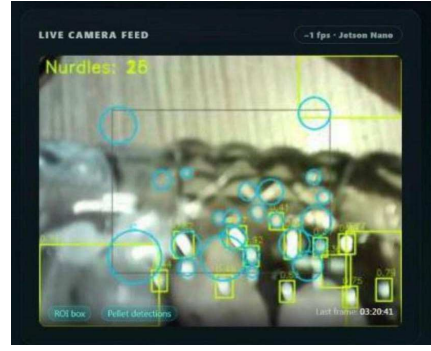


Figure 7: Representative detection visual outputs from the hybrid YOLOv8n + OpenCV pipeline.



Figure 8: Detection metrics from the hybrid YOLOv8n + OpenCV pipeline.

B. Edge Inference Efficiency

Translating the neural network model into ONNX format and running through onnxruntime decreased the required runtime memory usage by an order of magnitude (from 1.0 GB down to ~100 MB), necessary due to the limited shared memory available on the Jetson Nano of 4 GB. The CPU-based model inference managed more than 5 FPS on input size of 320×240 and exceeded the control loop frequency requirement of 5 Hz. The total end-to-end response latency of from the detection of the pellet until dashboard notification averaged 3.0 seconds, encompassing all stages (Figure 9).

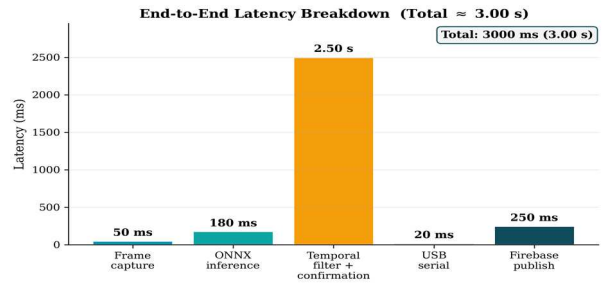


Figure 9: End-to-end latency breakdown. The temporal-filter confirmation window (2.5 s) dominates; inference and network transit contribute only ~430 ms in aggregate.

C. Multi-Sensor Fusion and False-Positive Suppression

With the four filters used, namely the ROI mask filter, circularity filter, EMA smoother, and confirmation timers, it was possible to achieve a successful elimination of any optical transient in the tests done on the benchtop. In the 30-minute-long test duration with the introduction of various perturbations (such as the presence of air bubbles, flickering of lights, and turbid water), the system only gave three false positive S2 (warning) alerts but zero S3 (alarms) prior to attaining steady state; there were no false S3 alerts at all due to just the air bubbles alone. This is since filters like these are

necessary based on previous studies done on AI-based optical microplastic sensor systems, which had shown that seawater factors besides microplastics were responsible for false positives [6]. In contrast, NurdleDNA reduces false positives with YOLO and OpenCV filters, turbidity detection, and FSM actuation.

Multi-sensor fusion lies at the heart of the actuation logic of NurdleDNA. Entry to S2 occurs via $LDR > 600$ OR AI count ≥ 4 , sustained for at least 1.5 seconds; entry to S3 via $MQ-135 > 200$ OR AI count ≥ 12 , sustained for at least 2.5 seconds. In this way, a graceful degradation scheme is provided: in the event of obstruction to the camera view or insufficient lighting, the analog sensors will ensure progression along the contamination route; conversely, failure of the gas sensor ensures continued control by the vision pipeline.

D. Deterministic Actuation and Containment Performance

The MG996R servo-actuated pinch valve seated within approximately 300 ms of the FSM entering S3 (command-to-seating). Because the FSM uses deterministic threshold logic, actuation is independent of Jetson Nano's availability. In a deliberate USB-serial disconnection test, the Arduino independently entered S3 in under 500 ms following a simultaneous LDR and MQ-135 excursion within one 5 Hz sample cycle, confirming the intended fail-operational behaviour. S3 is physically latched and can be cleared only by the operator through the reset button on pin D2.

Figures 10 and 11 illustrate the deterministic control strategy implemented in NurdleDNA. Figure 10 shows the time-aligned transition from WARN to ALARM, while Figure 11 presents the five-state Moore finite-state machine responsible for system actuation and latch-based alarm recovery.

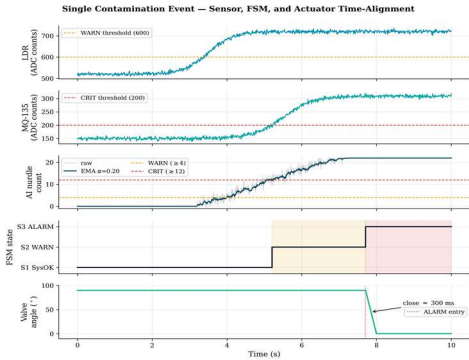


Figure 10: Time-aligned event trace showing WARN and CRIT transitions, ending with S3 activation and servo closure.

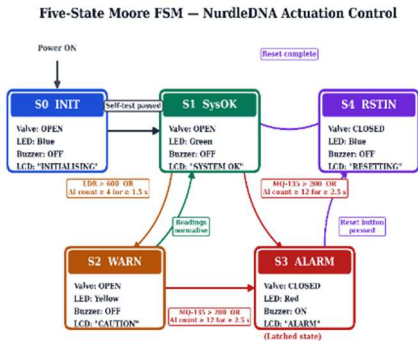


Figure 11: Five-state Moore finite-state machine governing actuation. S3 (ALARM) is physically latched and is cleared only by the operator reset on pin D2 (via state S4).

E. Evidence Capture and Gravimetric Logging

Using calibration against known reference masses via the HX711 `set_scale()` function, the load cell resolved the accumulated cartridge mass in grams at the Arduino's 5 Hz sampling rate (Fig. 12). During the live demonstration, the mass channel recorded approximately 2.4 g of captured material, which the dashboard displayed in real time alongside the AI density index.

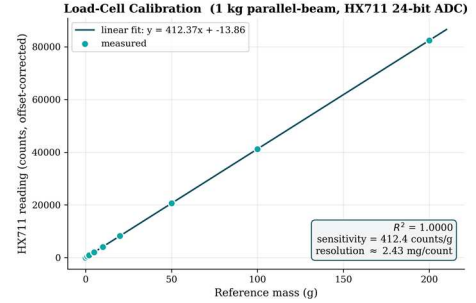


Figure 12: Load-cell calibration. Reference masses 0–200 g produce a linear response with $R^2 \approx 1.000$ and sensitivity ~ 412 counts/g (~ 2.4 mg per count).

F. Cloud Telemetry and Audit Logging

With the Firebase Realtime Database connection, 5 Hz sensor data recording and 1 Hz video stream (JPEG, 320x240 pixels, 60% compressed in base64) were possible as visualized in Figure 13. This combination of different frequencies made sure that total bandwidth usage would remain at around 200 kbps, which falls short of the LTE industrial gateway capacity, with 1 Hz video stream being responsible for most of the traffic (around 15 kB per JPEG, with base64 encoding increasing the size by 33%). Each change of state in ALARM was registered in the database along with time stamp, current state, sensor values and number of classes detected. It is this regulatory record that distinguishes NurdleDNA.



Figure 13: Operator dashboard rendered from the Firebase Realtime Database during an active S3 ALARM event.

G. System-Level Trade-offs and Limitations

Three system-level trade-offs and limitations warrant discussion.

First, the 2.5 s S3 confirmation window accounts for almost all of the end-to-end latency; shortening it would improve responsiveness at the cost of more false alarms and unnecessary valve closures. Second, the detector is an honest off-the-shelf baseline (overall $mAP@50 = 0.31$) limited by the public dataset's microscopy domain and severe class imbalance rather than by the system design; a curated nurdle/flow-cell dataset, class rebalancing (removing the dominant background class), and a larger YOLOv8s/m model on a Jetson Orin Nano are the clear routes to higher single-frame accuracy.

Third, the prototype is a single-channel inline unit; multi-channel bays would require multiple units or a manifold, although the Firebase schema already includes a bay_id field for this topology. Finally, several deployment-grade properties remain to be characterised: volumetric flow rate and capture efficiency, false-alarm and missed-detection rates over extended operation, and long-term reliability and maintenance intervals were not quantified in these bench trials and are required for field qualification. A formal ablation isolating the individual contributions of the YOLO detector, the OpenCV safety net, turbidity sensing, VOC sensing, and the FSM confirmation timers is likewise left to future work.

H. Comparison with Existing Approaches

The comparison between NurdleDNA and traditional monitoring techniques is provided in Table I. Both FTIR and Raman spectroscopy have high chemical specificity, but these techniques work off-line. The turbidity sensors and particle counters cannot differentiate nurdles from bubbles, sand, and bio-particles. The passive filtration technique can document pollution only, while NurdleDNA can document and physically arrest it.

TABLE I. COMPARISON OF NURDLE DNA WITH CONVENTIONAL MICROPLASTIC-MONITORING APPROACHES.

Approach	Real-time	Source-level containment	Forensic Evidence	Chemical Specificity	Field deployable	Cost (reL)
Manual inspection [3]	No	No	Photo only	Low	Yes	Low
Passive Filtration [1],[3]	No	Partly	Bulk only	None	Yes	Low
Turbidity/particle counter [6],[7]	Yes	No	Indirect	None	Yes	Low
FTIR spectroscopy [3]	No	No	Yes	High	No	High
Raman spectroscopy [3]	Partly	No	Yes	High	Partly	High
VIS-NIR + chemometrics [7]	Partly	No	Yes	Medium	Partly	Medium
Nurdle-DNA (proposed system)	Yes (~3s)	Yes (valve)	3-channel	Medium	Yes	Medium

V. CONCLUSION AND FUTURE WORK

This paper outlined the design of NurdleDNA, which is an inline AI and IoT solution to a particular problem in industrial

water monitoring: the lack of a means to intercept, in real time, the pre-production plastic pellets at source level. This is achieved through the integration of YOLOv8n object detector on NVIDIA Jetson Nano with Moore FSM determinism on Arduino, multichannel fusion involving optical, gas, and gravimetric sensor input, and Firebase-driven cloud-based audit log.

Three areas of potential research going forward are suggested. (1) Hardening for the field environment and multi-bay operations, necessitating IP-rated enclosures and SCADA compatibility; the Firebase bay_id architecture already addresses this point. (2) Better classification of fragments through an increased dataset of weathered samples combined with the use of the YOLOv8s/m architecture on the Jetson Orin Nano platform. (3) Determination of the polymer type via miniature Raman or VIS-NIR spectrometry at the evidence cartridge stage to complete the “nurdle DNA” paradigm.

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